Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · AI/ML · LangGraph · RAG Pipelines · LLM Evaluation · FastAPI · Python · TypeScript · React · AWS · GCP · PostgreSQL · Docker · 9yr Production · Multifamily Real Estate

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PROFESSIONAL SUMMARY

Senior software engineer with 9+ years of production experience across full-stack development, AI/ML systems, data pipelines, and regulated compliance engineering. At Entrata I owned mission-critical Python backend systems for 7 years at 99.5% uptime - HUD affordable housing compliance pipelines, async distributed processing, and AWS data infrastructure serving 100K+ users and 1M+ property records with zero compliance violations. Currently building AI-native systems: AgentProbe (open-source AI evaluation toolkit - LangGraph, FastAPI, ChromaDB RAG, CI regression gate, 5-metric quality engine) and Leasel (8-agent LangGraph platform with TF-IDF RAG, HITL guardrails, and 93.6% precision across 1,240 compliance test cases). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Based in Arlington, VA.

TECHNICAL SKILLS

Languages: Python (9yr), PHP, JavaScript, TypeScript, SQL, PL/SQL, R, Bash, HTML/CSS, jQuery, AJAX

AI & Agents: LangChain, LangGraph, Agentic AI, Multi-Agent Orchestration, LLM Orchestration, Tool Calling, ChromaDB, Vector Databases, RAG Pipelines, Semantic Search, Embeddings, NLP, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Model Evaluation, LLMOps, Agent Observability, Agent Tracing, Prompt Engineering, Few-Shot Prompting, Chain-of-Thought, Output Validation, Context Engineering, OCR, Document Intelligence, HITL, Human-in-the-Loop, Confidence Scoring, Guardrails, Production AI, OpenAI API, Anthropic API, Groq, Ollama, FastAPI, REST APIs, MCP

ML & Data: PyTorch, TensorFlow, XGBoost, scikit-learn, Isolation Forest, Random Forest, Logistic Regression, Anomaly Detection, Feature Engineering, Statistical Modeling, Model Deployment, NumPy, Pandas, Matplotlib, Seaborn, Plotly, spaCy, BERT, Hugging Face, Jupyter, Data Visualization, Tableau, Power BI, ggplot2

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue, Lambda), GCP (Cloud Storage), Docker, CI/CD (GitHub Actions), Apache Spark, PySpark, RabbitMQ, Redis, Microservices, Distributed Systems, Observability, Performance Engineering, ETL, Data Pipelines, Data Modeling, PostgreSQL, MySQL, Oracle, NoSQL, SQLite, Libsodium, Git, Linux, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer (Unpaid)

- Designed and deployed a production agentic AI system (LangGraph stateful graph: parse → plan → retrieve → validate → conditional routing) as sole technical owner - delivered risk-aware compliance strategy: compliant missions route to export, non-compliant ones route to rejection with grounded reasons.
- Parsed natural-language survey requests into structured flight plans via quantitative photogrammetry math (ground sampling distance, camera footprint, overlap-driven line spacing); implemented RAG layer over FAA Part 107 regulatory corpus with cited outputs.
- Engineered validation guardrails enforcing FAA altitude limits, battery budgets, and geometric no-fly-zone checks; built evaluation pipeline, CI validation gate, HITL guardrails, and observability infrastructure; delivered 0→1 in weeks.

George Mason University - College of Engineering and Computing
Aug 2024 - May 2026

Graduate Teaching Assistant (Aug 2025 - May 2026, Part-time) & Research Associate (Aug 2024 - Aug 2025)

- Built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets across 190+ countries - R^2=0.977 regression accuracy; designed full pipeline from data collection and transformation through hypothesis testing and validation on AWS and GCP infrastructure.
- Applied Isolation Forest to automate data integrity checks across 3 research pipelines, surfacing anomalies that manual review missed and improving downstream model reliability.
- Designed Tableau, ggplot2, and Quarto dashboards for faculty leadership - improved research comprehension 40% across two cohorts.
- Delivered technical workshops to 600+ students across multiple sections of MIS 303: Python programming (data structures, OOP, automation scripting) and database design (ER modeling, SQL, MS Access/Excel); provided one-on-one debugging support and graded assignments with rubric-aligned feedback.

Entrata - PropTech / Multifamily Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **HUD compliance & regulatory data pipelines** - owned the annual HUD income-limit import pipeline for nearly 7 years: few thousand federal source rows expanded into ~180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units; every release included baseline validation, reconciliation, and rollback preparation because one wrong import means thousands of residents get the wrong eligibility decision; zero compliance violations across full ownership period.
- **Async distributed systems** - built and maintained RabbitMQ-based async processing system for gross-rent updates across thousands of tenant records; designed retry logic and idempotency guarantees eliminating double-processing failures; owned through production incidents and annual rent-schedule changes.
- **Database performance & optimization** - optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, partitioning, and stored procedure refactoring; owned complex schemas for eligibility, financial, and document workflows across multi-tenant production environments.
- **AWS cloud & data migration** - led Affordable Housing module integration for AWS S3 document migration; inventoried all document types, mapped legacy records to framework contracts, validated migrated content and metadata; achieved 100% PHI compliance with Libsodium-encrypted pipelines (zero security incidents across entire resident data migration).
- **Full-stack production delivery** - shipped 20+ production modules (PHP, Python, PostgreSQL, JavaScript, React.js, TypeScript, RabbitMQ) across 120+ Agile sprints; led Docker microservices redesign (50% scalability improvement, 30% deployment time reduction); GitHub Actions CI/CD with automated unit, performance, and regression testing; eliminated 80% of manual operational workload via Python/Bash automation pipelines.
- **Observability & incident response** - built structured logging, monitoring, and alerting infrastructure; authored runbooks; led incident response with RCA; built internal debugging tools reducing triage time 40%; kept bug count below 10 across 20+ production modules over 120+ sprints.
- **Technical leadership** - wrote design documents and architecture decisions for 20+ production modules; mentored 5+ engineers on system design and production practices; 300+ peer code reviews; built consensus across product, compliance, and engineering stakeholders.

Entrata India (Db Xento Systems) - Pune

Jan 2017 - Sep 2018

Associate Software Engineer (Jul 2017 - Sep 2018) → Trainee (Jan 2017 - Jun 2017)

- Built property management modules (PHP, Python, PostgreSQL) for affordable housing and rental management, including income limits import (1.6M HUD/Tax Credit records across 5 years), document management, and compliance workflows.
- Led cloud migration of client documents from legacy server storage to AWS S3 - resolved 90% of legacy space constraints; created 56 state-specific housing certification documents with merge fields.
- Redesigned outdated document storage database architecture, migrating hundreds of thousands of records to a modern schema; provided 6 months of post-deployment stabilization.
- Acquired foundational skills in PHP, PostgreSQL, PL/SQL, and Agile practices; contributed to affordable housing data workflows under senior developer guidance.

PROJECTS

Leasel: Agentic AI Support Investigation Platform

LangGraph · RAG · HITL · FastAPI · MCP · Multi-Agent · Production

Python · LangGraph (8-agent StateGraph) · FastAPI · TF-IDF RAG · OCR (pdfplumber, pytesseract) · SQLite · MCP · Anthropic API · Streamlit | github.com/venkatalakshmiparimala/Lease_complaince_rag

Three-surface enterprise AI knowledge platform sharing one SQLite evidence model: (1) **Document Intelligence & Review** - OCR-fallback ingestion, section-aware chunking, TF-IDF retrieval, 100% citation correctness, missing/conflicting clause detection, human-review gate on regulated decisions; (2) **Support Ticket Investigation** - 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) with conditional routing, tool calling, and MCP integration for live HUD income-limit API; (3) **Tenant Self-Service** - policy questions answered directly; eligibility questions always route to staff, never resolved by AI. Core design rule enforced in code: LLM extracts and reasons; human or deterministic logic makes every final compliance call. **Measured outcomes:** 93.6% precision, 87.9% F1, pass@1 81.7% / pass@3 88.3% across 1,240 residential lease PDFs (5 US states + federal, 1,050 synthetic tenants).

AgentProbe: Open-Source AI Evaluation Toolkit

LLM Evaluation · LangGraph · ChromaDB · FastAPI · CI Gate · Observability · Open Source

Python · FastAPI · LangGraph · React.js · TypeScript · ChromaDB · GitHub Actions · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production AI evaluation platform: 15-endpoint FastAPI backend, React.js + TypeScript frontend, multi-provider LLM routing (Anthropic, OpenAI, Ollama). **Evaluation engine:** 5-metric quality measurement (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-Judge scoring calibrated against expected outputs; pass@k multi-trial harness surfacing variance in non-deterministic AI outputs. **RAG pipeline:** ChromaDB semantic retrieval with embedding-based search; retrieval quality scored independently from generation quality. **CI regression gate:** GitHub Actions integration blocks deployment when quality metrics regress - zero regressions across 1,240+ test cases. **Observability:** structured distributed tracing across all agent steps; monitoring and alerting on quality drift. Fully local, zero vendor dependency, open-source.

AI on FHIR: NLP-Driven Medical Records Query System

Full Stack · AWS Lambda · HIPAA · LLM · React · TypeScript · BERT · Docker

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · PostgreSQL · REST APIs · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI application for a regulated healthcare environment: React.js + TypeScript frontend, Python/Flask REST API backend, LLM-powered NLP translating natural-language queries (EN/ES) into structured FHIR API calls using LLMs, BERT, and embeddings. AWS Lambda serverless backend with API gateway; HIPAA-compliant RBAC with full audit trail on every AI-assisted retrieval; PostgreSQL data services; Docker containerization; complete observability stack. ~60% reduction in manual clinical retrieval time.

CrashClarity: Large-Scale Road Accident Analysis - 104K Records

Random Forest · Logistic Regression · Feature Engineering · Python · R · scikit-learn

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages | Portfolio

End-to-end ML pipeline on 104K+ UK road accident records: data ingestion and normalization, feature engineering across weather, road surface, temporal, and geographic variables, Random Forest + Logistic Regression ensemble (85%+ accuracy), cross-validation, and interactive dashboard deployed on GitHub Pages. Demonstrates large-scale data analysis, predictive modeling, and data-driven insight communication.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Fairfax, Virginia — May 2026 · GPA 3.87

Coursework: Machine Learning, Deep Learning, Data Mining, Statistical Modeling, Cloud Computing, Database Systems, Applied Analytics

B.Tech Computer Science & Engineering - Jawaharlal Nehru Technological University, Anantapur — 2013-2017

Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Software Engineering, OOP

VOLUNTEER & COMMUNITY

Saving Hands NGO - Pune, India — 2018 - 2024

Volunteer Educator & Support Worker (concurrent with full-time employment)

- Provided one-on-one academic support to visually impaired children over 6 years - teaching core subjects, assisting with exam preparation, and adapting instructional approaches to individual learning needs.
- Conducted health checkups and participated in fundraising to support program operations and expand community outreach.